PRG 1 - Website creation using html and css tha 1st program

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Webpage</title>
</head>
<body>

  <h1>Welcome to My Webpage</h1>
  <p>This is a simple HTML webpage created for MCA students.</p>

  <h2>About Me</h2>
  <p>I love learning web technologies like HTML, CSS, and JavaScript.</p>

  <h3>My Hobbies</h3>
  <ul>
    <li>Reading</li>
    <li>Coding</li>
    <li>Music</li>
  </ul>

  <img src="https://via.placeholder.com/250" alt="Sample Image">

  <br><br>
  <a href="https://www.google.com">Visit Google</a>

</body>
```

</html>

PROGRAM 2: Personal Portfolio Webpage using HTML and CSS

File 1: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Personal Portfolio</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <h1>My Portfolio</h1>
    <nav>
      <a href="#about">About</a>
      <a href="#skills">Skills</a>
      <a href="#projects">Projects</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <section id="about">
    <h2>About Me</h2>
```

```
        <p>Hello! I am a web developer passionate about designing modern and responsive websites.</p>
    </section>

    <section id="skills">
        <h2>My Skills</h2>
        <ul>
            <li>HTML & CSS</li>
            <li>JavaScript</li>
            <li>Python</li>
            <li>React</li>
        </ul>
    </section>

    <section id="projects">
        <h2>Projects</h2>
        <div class="project-box">
            <h3>Project 1</h3>
            <p>A responsive website created using HTML, CSS, and JS.</p>
        </div>
        <div class="project-box">
            <h3>Project 2</h3>
            <p>Online portfolio site with modern UI.</p>
        </div>
    </section>

    <footer id="contact">
        <h2>Contact Me</h2>
```

```
        <p>Email: myemail@example.com</p>
    </footer>

</body>
</html>
```

File 2: style.css

```
body {
    font-family: Arial, sans-serif;
    margin: 0;
    padding: 0;
    background: #f2f2f2;
}

header {
    background: #333;
    color: white;
    padding: 20px;
    text-align: center;
}

nav a {
    color: #f7c08a;
    margin: 0 15px;
    text-decoration: none;
    font-weight: bold;
}

section {
```

```
    padding: 20px;
    background: white;
    margin: 20px;
    border-radius: 10px;
}

.project-box {
    background: #e8e8e8;
    padding: 10px;
    margin: 10px 0;
    border-radius: 8px;
}
footer {
    background: #333;
    color: white;
    padding: 20px;
    text-align: center;
}
```

PROGRAM 3: Designing an HTML Page with Frames

frameset.html

```
<!DOCTYPE html>
<html>
<head>
   <title>HTML Frames Example</title>
</head>
<frameset rows="20%,*">
   <frame src="header.html" name="headerFrame">
   <frameset cols="25%,75%">
      <frame src="menu.html" name="menuFrame">
      <frame src="content.html" name="contentFrame">
   </frameset>
</frameset>
</html>
```

header.html

```
<!DOCTYPE html>
<html>
<body style="background-color: #333; color: white; text-align: center; padding:
10px;">
   <h1>Welcome to My Website</h1>
</body>
</html>
```

menu.html

```
<!DOCTYPE html>
<html>
<body style="background-color: #f0f0f0;">
```

```
    <h3>Menu</h3>
    <a href="content.html" target="contentFrame">Home</a><br>
    <a href="about.html" target="contentFrame">About</a>
</body>
</html>
```

content.html

```
<!DOCTYPE html>
<html>
<body>
    <h2>This is the main content area.</h2>
    <p>Select an option from the menu.</p>
</body>
</html>
```

PROGRAM 4

Build a Chat module using HTML CSS and JavaScript

HTML File

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Chat Module</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="chat-container">
    <div class="chat-header">Chat Module</div>

    <div class="chat-box" id="chatBox"></div>

    <div class="chat-input">
        <input type="text" id="userInput" placeholder="Type a message..." />
        <button id="sendBtn">Send</button>
    </div>
</div>

<script src="script.js"></script>
</body>
</html>
```

CSS FILE

```
body {
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
}

.chat-container {
    width: 350px;
    margin: 50px auto;
    background: white;
    border-radius: 8px;
    box-shadow: 0 0 10px rgba(0,0,0,0.1);
}

.chat-header {
    background: #0078d7;
    color: white;
    padding: 10px;
    text-align: center;
    font-weight: bold;
}

.chat-box {
    height: 300px;
    padding: 10px;
    overflow-y: auto;
    display: flex;
    flex-direction: column;
}
```

```
.message {
    margin: 6px 0;
    padding: 8px 12px;
    border-radius: 6px;
    max-width: 75%;
}

.user {
    background: #dcf8c6;
    align-self: flex-end;
}

.bot {
    background: #ececec;
    align-self: flex-start;
}

.chat-input {
    display: flex;
    padding: 10px;
}

.chat-input input {
    flex: 1;
    padding: 8px;
}

.chat-input button {
```

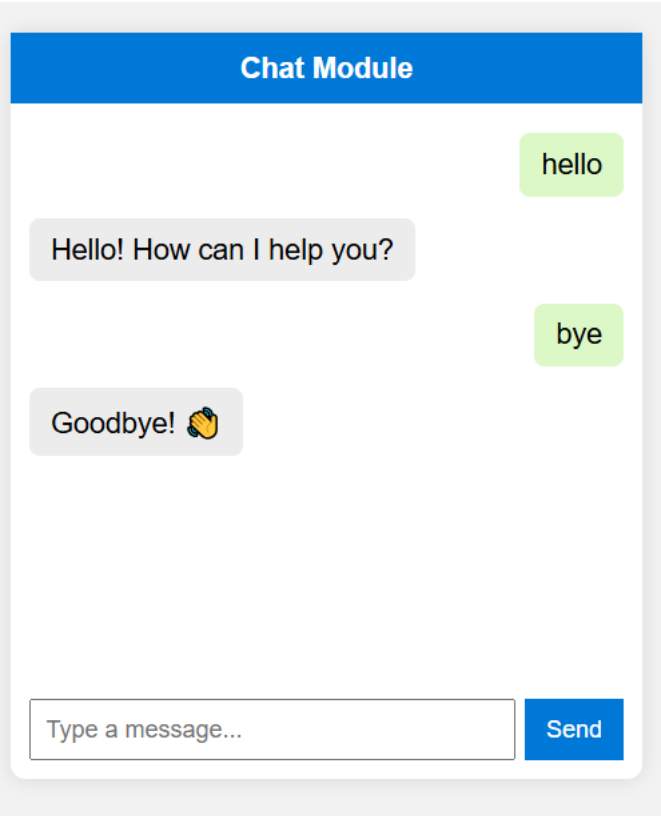
```
    margin-left: 5px;
    padding: 8px 12px;
    background: #0078d7;
    color: white;
    border: none;
    cursor: pointer;
}
```

JAVASCRIPT FILE

```
const input = document.getElementById("userInput");
const chatBox = document.getElementById("chatBox");
const sendBtn = document.getElementById("sendBtn");

sendBtn.addEventListener("click", sendMessage);

input.addEventListener("keypress", function (e) {
    if (e.key === "Enter") {
        sendMessage();
    }
});

function sendMessage() {
    const message = input.value.trim();
    if (message === "") return;

    addMessage(message, "user");
    input.value = "";
```

```
    setTimeout(() => {
        botReply(message);
    }, 700);
}

function addMessage(text, sender) {
    const msg = document.createElement("div");
    msg.className = `message ${sender}`;
    msg.textContent = text;
    chatBox.appendChild(msg);
    chatBox.scrollTop = chatBox.scrollHeight;
}

function botReply(userText) {
    let reply = "I am a simple chatbot 🤖";

    if (userText.toLowerCase().includes("hello"))
        reply = "Hello! How can I help you?";
    else if (userText.toLowerCase().includes("bye"))
        reply = "Goodbye! 👋";

    addMessage(reply, "bot");
}
```

OUTPUT


Chat Module
hello
Hello! How can I help you?
bye
Goodbye! 👋
Type a message...
Send

PROGRAM 5

Create a simple calculator Application using React JS

Index.js

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import "./styles.css";

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<App />);
```

App.js

```
import React, { useState } from "react";

function App() {
  const [input, setInput] = useState("");

  const handleClick = (value) => {
    setInput(input + value);
  };

  const calculateResult = () => {
    try {
      setInput(eval(input).toString());
    } catch {
      setInput("Error");
    }
```

```
};

const clearInput = () => {
  setInput("");
};

return (
  <div className="calculator">
    <h2>React Calculator</h2>

    <input type="text" value={input} readOnly />

    <div className="buttons">
      <button onClick={() => handleClick("7")}>7</button>
      <button onClick={() => handleClick("8")}>8</button>
      <button onClick={() => handleClick("9")}>9</button>
      <button onClick={() => handleClick("/")}>/</button>

      <button onClick={() => handleClick("4")}>4</button>
      <button onClick={() => handleClick("5")}>5</button>
      <button onClick={() => handleClick("6")}>6</button>
      <button onClick={() => handleClick("*")}>*</button>

      <button onClick={() => handleClick("1")}>1</button>
      <button onClick={() => handleClick("2")}>2</button>
      <button onClick={() => handleClick("3")}>3</button>
      <button onClick={() => handleClick("-")}>-</button>

      <button onClick={() => handleClick("0")}>0</button>
```

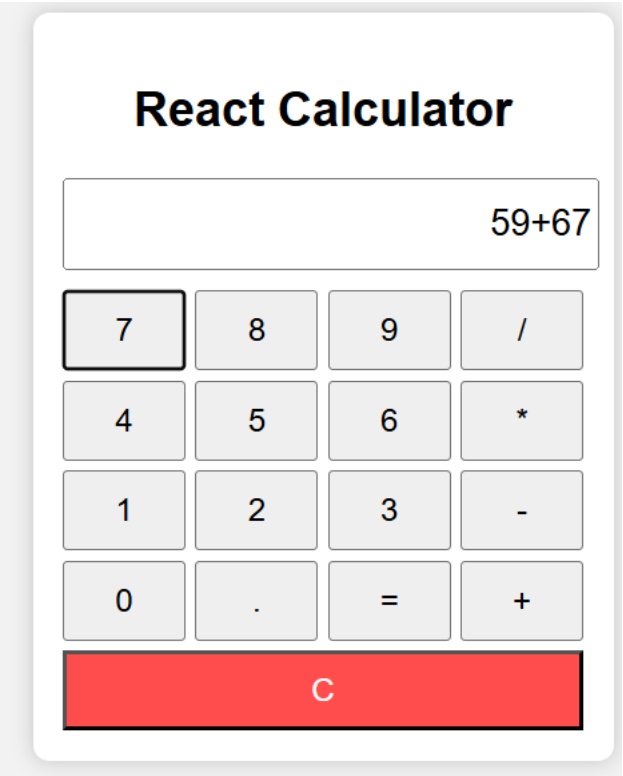
```
        <button onClick={() => handleClick(".")}>.</button>
        <button onClick={calculateResult}>=</button>
        <button onClick={() => handleClick("+")}>+</button>

        <button className="clear" onClick={clearInput}>C</button>
      </div>
    </div>
  );
}

export default App;
```

styles.css

```
body {
  font-family: Arial, sans-serif;
  background: #f2f2f2;
}

.calculator {
  width: 260px;
  margin: 60px auto;
  background: white;
  padding: 15px;
  border-radius: 8px;
  box-shadow: 0 0 10px rgba(0,0,0,0.2);
  text-align: center;
}

.calculator input {
```

```
  width: 100%;
  height: 40px;
  margin-bottom: 10px;
  font-size: 18px;
  text-align: right;
}

.buttons {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 5px;
}

button {
  height: 40px;
  font-size: 16px;
  cursor: pointer;
}

.clear {
  grid-column: span 4;
  background: #ff4d4d;
  color: white;
}
```

Output


React Calculator
59+67
7
8
9
/
4
5
6
*
1
2
3
-
0
.
=
+
C

PROGRAM 6

Create a voting application using React JS

index.js

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import "./styles.css";

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<App />);
```

App.js

```
import React, { useState } from "react";

function App() {
  const [votes, setVotes] = useState({
    CandidateA: 0,
    CandidateB: 0,
    CandidateC: 0,
  });

  const vote = (candidate) => {
    setVotes({
      ...votes,
      [candidate]: votes[candidate] + 1,
    });
  };

  const resetVotes = () => {
```

```
  setVotes({
    CandidateA: 0,
    CandidateB: 0,
    CandidateC: 0,
  });
};

return (
  <div className="app">
    <h2>Voting Application</h2>

    <div className="candidate">
      <h3>Candidate A</h3>
      <p>Votes: {votes.CandidateA}</p>
      <button onClick={() => vote("CandidateA")}>Vote</button>
    </div>

    <div className="candidate">
      <h3>Candidate B</h3>
      <p>Votes: {votes.CandidateB}</p>
      <button onClick={() => vote("CandidateB")}>Vote</button>
    </div>

    <div className="candidate">
      <h3>Candidate C</h3>
      <p>Votes: {votes.CandidateC}</p>
      <button onClick={() => vote("CandidateC")}>Vote</button>
    </div>
```

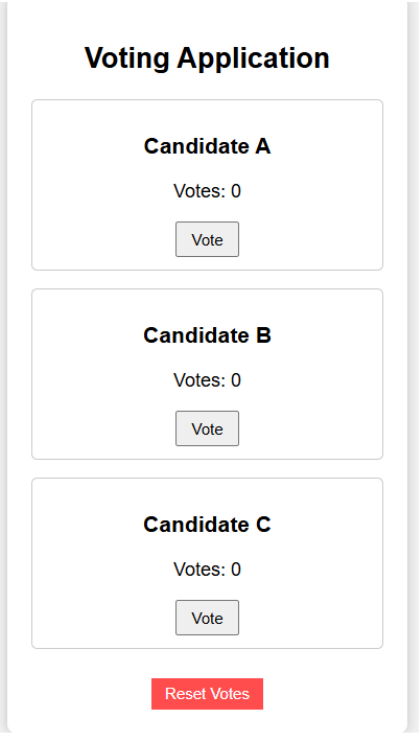
```
      <button className="reset" onClick={resetVotes}>Reset Votes</button>
    </div>
  );
}

export default App;
```

App.css

```
body {
  font-family: Arial, sans-serif;
  background-color: #f2f2f2;
}

.app {
  width: 300px;
  margin: 50px auto;
  background: white;
  padding: 20px;
  border-radius: 8px;
  box-shadow: 0 0 10px rgba(0,0,0,0.2);
  text-align: center;
}

.candidate {
  margin-bottom: 15px;
  padding: 10px;
  border: 1px solid #ccc;
  border-radius: 5px;
}
```

```
button {
  padding: 6px 12px;
  cursor: pointer;
}

.reset {
  margin-top: 10px;
  background-color: #ff4d4d;
  color: white;
  border: none;
}
```

OUTPUT



PROGRAM 7

PHP-MySQL – Insert Form Data into Database

MySQL:

```
CREATE DATABASE student_db;
USE student_db;
CREATE TABLE students (
   id INT AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(50),
   email VARCHAR(50),
   course VARCHAR(50)
);
```

Index HTML

```
<!DOCTYPE html>
<html>
<head>
<title>Student Registration</title>
</head>
<body>
        <h2>Student Registration Form</h2>
        <form action="insert.php" method="post">
Name:<input type="text" name="name" required><br><br>
Email:<input type="email" name="email" required><br><br>
Course:<input type="text" name="course" required><br><br>
<input type="submit" value="Submit">
</form>
</body>
</html>
```

PHP:

```php
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "student_db";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
// Get form data
$name = $_POST['name'];
$email = $_POST['email'];
$course = $_POST['course'];
// Insert query
$sql = "INSERT INTO students (name, email, course)
VALUES ('$name', '$email', '$course')";

if ($conn->query($sql) === TRUE) {
    echo "New record inserted successfully";
} else {
    echo "Error: " . $conn->error;
}
$conn->close();
?>
```

OUTPUT

Student Registration Form

Name: Ramesh

Email: ramesh@gmail.com

Course: BCA

Submit

New record inserted successfully

students

id		name	email	course
1	1	Ramesh	ramesh@gmail.com	BCA
2	2	Priya	priya@gmail.com	BSc

PROGRAM 8

Create a Dynamic Clock using JavaScript DOM

HTML + JavaScript Program

```
<!DOCTYPE html>
<html>
<head>
<title>Dynamic Digital Clock</title>
<style>
body{
   text-align:center;
   font-family: Arial;
}
#clock{
   font-size:40px;
   color:blue;
   margin-top:100px;
}
</style>
</head>
<body>
<h2>Digital Clock</h2>
<div id="clock"></div>
<script>
function showTime(){
   var date = new Date();
   var h = date.getHours();
   var m = date.getMinutes();
   var s = date.getSeconds();
   var time = h + ":" + m + ":" + s;
```

```
    document.getElementById("clock").innerHTML = time;
}
setInterval(showTime,1000);
</script>
</body>
</html>
```

OUTPUT

PROGRAM 9

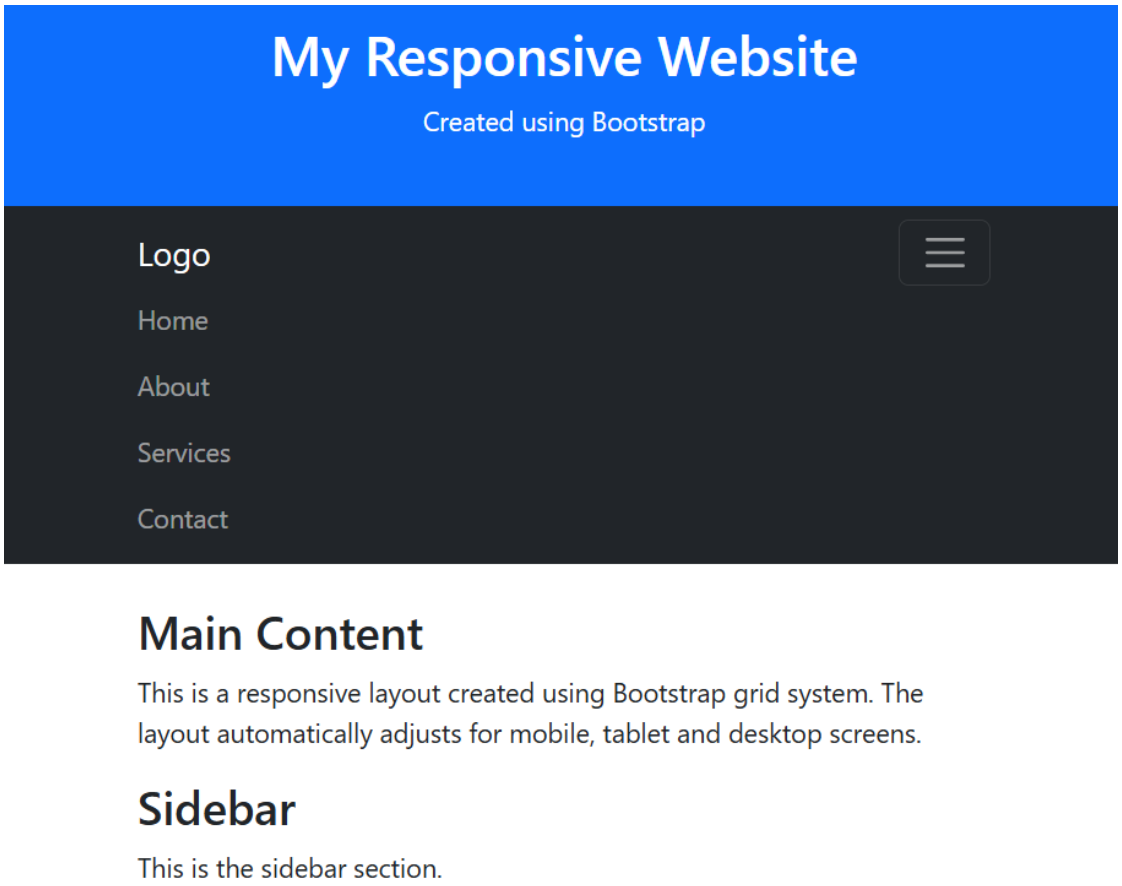
Create a Responsive Website Layout using Bootstrap

```
<!DOCTYPE html>

<html>

<head>

<title>Responsive Website using Bootstrap</title>

<!-- Bootstrap CSS -->

<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css">

</head>

<body>

<!-- Header -->

<div class="container-fluid bg-primary text-white text-center p-4">

<h1>My Responsive Website</h1>

<p>Created using Bootstrap</p>

</div>

<!-- Navigation Menu -->

<nav class="navbar navbar-expand-lg navbar-dark bg-dark">

<div class="container">

<a class="navbar-brand" href="#">Logo</a>

<button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#menu">

<span class="navbar-toggler-icon"></span>

</button>

<div class="collapse navbar-collapse" id="menu">

<ul class="navbar-nav ms-auto">

<li class="nav-item">

<a class="nav-link" href="#">Home</a>

</li>

<li class="nav-item">
```

```
<a class="nav-link" href="#">About</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Services</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Contact</a>
</li>
</ul>
</div>
</div>
</nav>
<!-- Main Content -->
<div class="container mt-4">
<div class="row">
<div class="col-md-8">
<h2>Main Content</h2>
<p>
This is a responsive layout created using Bootstrap grid system.
The layout automatically adjusts for mobile, tablet and desktop screens.
</p>
</div>
<div class="col-md-4">
<h2>Sidebar</h2>
<p>This is the sidebar section.</p>
</div>
</div>
</div>
<!-- Footer -->
```

```
<div class="container-fluid bg-dark text-white text-center p-3 mt-4">
<p>© 2026 My Website</p>
</div>
<!-- Bootstrap JS -->
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></s
cript>
</body>
</html>
```

OUTPUT



1. Create MySQL Database

```
CREATE DATABASE login_system;

USE login_system;

CREATE TABLE users (

  id INT AUTO_INCREMENT PRIMARY KEY,

  username VARCHAR(100),

  email VARCHAR(100),

  password VARCHAR(255)

);
```

2. Database Connection (db.php)

```
<?php

$conn = new mysqli("localhost","root","","login_system");

if($conn->connect_error)

{

  die("Connection Failed: " . $conn->connect_error);

}

?>
```

3. User Registration Page (register.php)

```php
<?php

include "db.php";

if(isset($_POST['register']))

{

  $username = $_POST['username'];

  $email = $_POST['email'];

  $password = password_hash($_POST['password'], PASSWORD_DEFAULT);

  $sql = "INSERT INTO users(username,email,password)

       VALUES('$username','$email','$password')";

  if($conn->query($sql)==TRUE)

  {

     echo "Registration Successful";

  }

  else

  {

     echo "Error";

  }
```

```
}

?>

<h2>User Registration</h2>

<form method="POST">

Username: <input type="text" name="username"><br><br>

Email: <input type="email" name="email"><br><br>

Password: <input type="password" name="password"><br><br>

<input type="submit" name="register" value="Register">

</form>

<a href="login.php">Go to Login</a>
```

4. Login Page (login.php)

```
<?php

session_start();

include "db.php";

if(isset($_POST['login']))

{

  $email = $_POST['email'];

  $password = $_POST['password'];
```

```php
$sql = "SELECT * FROM users WHERE email='$email'";

$result = $conn->query($sql);

if($result->num_rows > 0)

{

    $row = $result->fetch_assoc();

    if(password_verify($password,$row['password']))

    {

        $_SESSION['username'] = $row['username'];

        header("Location: dashboard.php");

    }

    else

    {

        echo "Invalid Password";

    }

}

else

{

    echo "User not found";

}
```

```
}

?>

<h2>Login</h2>

<form method="POST">

Email: <input type="email" name="email"><br><br>

Password: <input type="password" name="password"><br><br>

<input type="submit" name="login" value="Login">

</form>

<a href="register.php">Create Account</a>
```